

Netflix_eda

June 22, 2026

```
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[11]: df = pd.read_csv('Netflix_Movies_and_TV_Shows_cleaned.csv')
```

```
[88]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 8797 entries, 0 to 8796
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   show_id         8797 non-null   str
 1   type            8797 non-null   str
 2   title           8797 non-null   str
 3   director        8797 non-null   str
 4   cast            8797 non-null   str
 5   country         8797 non-null   str
 6   date_added      8709 non-null   str
 7   release_year    8797 non-null   int64
 8   rating          8797 non-null   str
 9   duration        8797 non-null   str
10   listed_in       8797 non-null   str
11   description     8797 non-null   str
dtypes: int64(1), str(11)
memory usage: 3.8 MB
```

1 Observation:

- The dataset contains X rows and Y columns.
- Most columns have appropriate data types.
- Few/no missing values are present.

```
[21]: df.describe()
```

```
[21]:      release_year
      count    8797.000000
      mean     2014.183472
      std       8.822191
      min      1925.000000
      25%      2013.000000
      50%      2017.000000
      75%      2019.000000
      max      2021.000000
```

2 Observation:

- The numerical features show variation in their values.
- The mean and median indicate the overall distribution of the data.

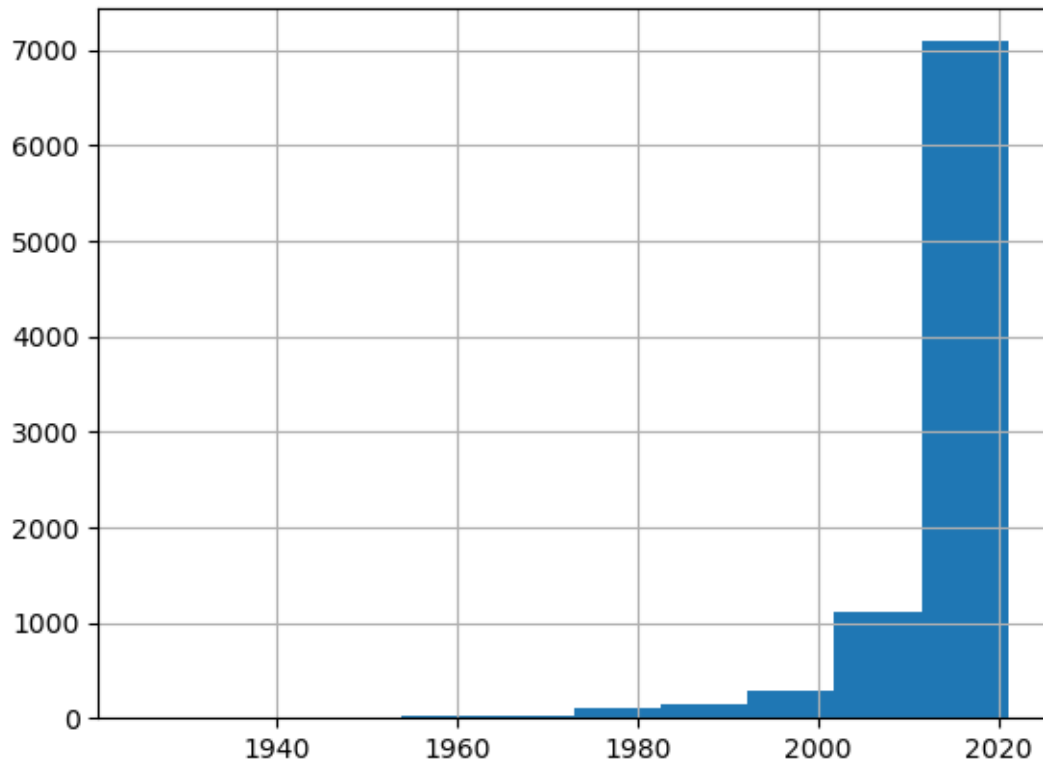
```
[41]: df['type'].value_counts()
```

```
[41]: type
      Movie      6131
      TV Show    2666
      Name: count, dtype: int64
```

3 Observation:

- Movies are more common than TV Shows.
- Some categories appear much more frequently than others.

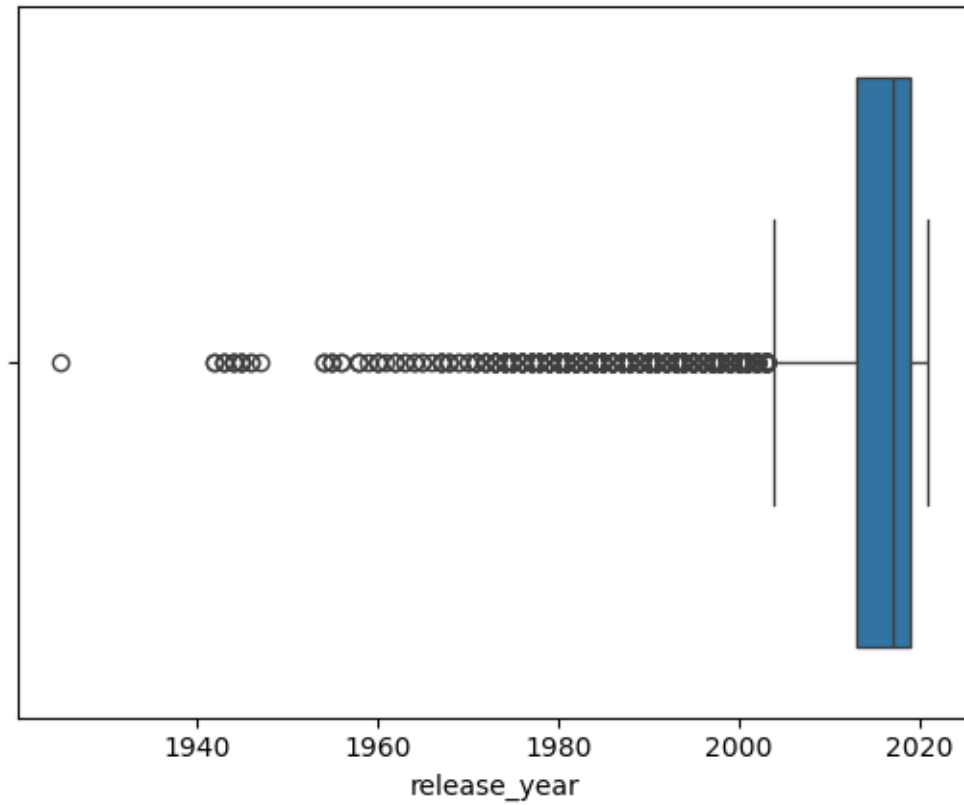
```
[83]: df['release_year'].hist()
      plt.show()
```



4 Observation:

- Most content was released in recent years.
- The distribution is concentrated after 2010.

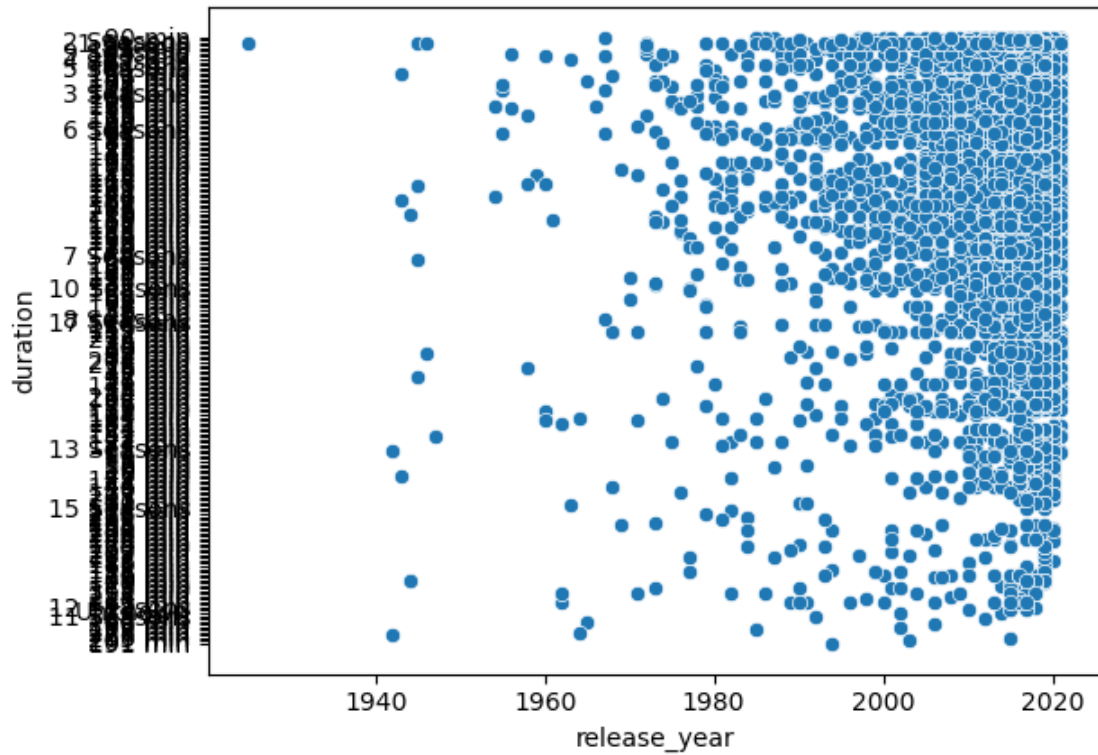
```
[80]: sns.boxplot(x=df['release_year'])  
plt.show()
```



5 Observation:

- A few outliers are visible in the dataset.
- Most values lie within a limited range.

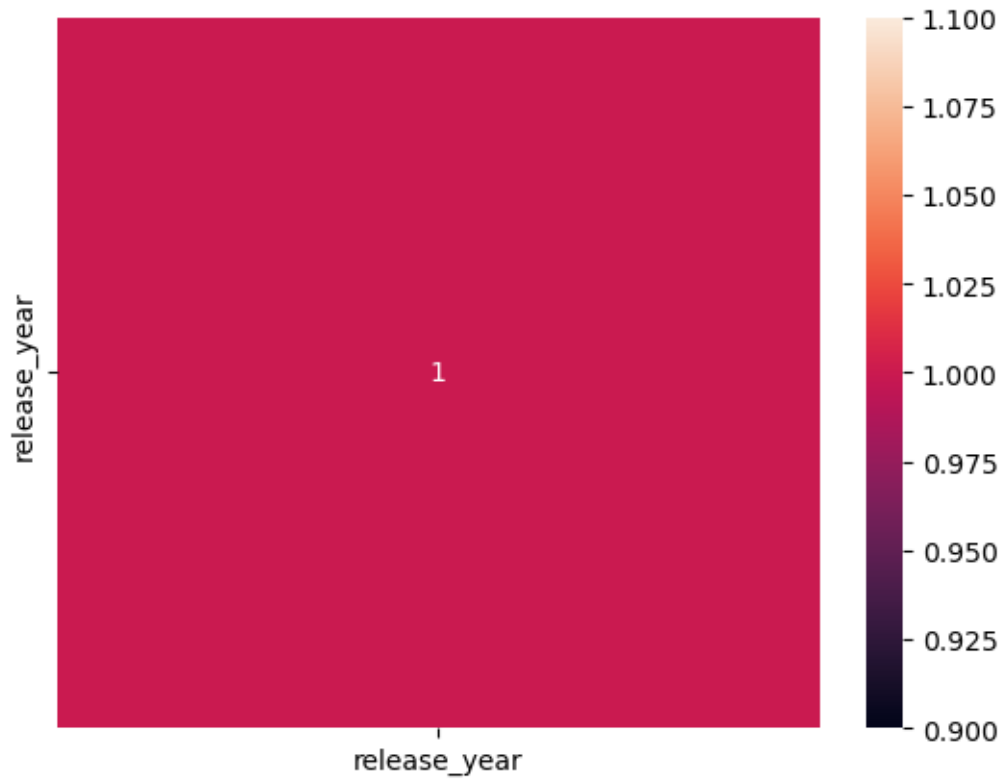
```
[57]: sns.scatterplot(x='release_year', y='duration', data=df)  
plt.show()
```



6 Observation:

- No strong relationship is observed between the selected variables.
- The points are widely scattered.

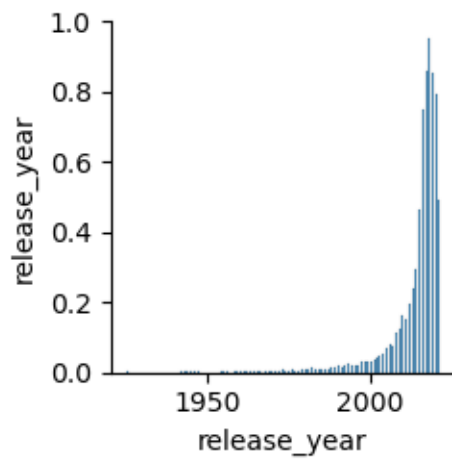
```
[76]: sns.heatmap(df.select_dtypes(include='number').corr(), annot=True)
plt.show()
```



7 Observation:

- Most numerical variables have weak correlations.
- No strong positive or negative correlation is present.

```
[78]: sns.pairplot(df.select_dtypes(include='number'))  
plt.show()
```



8 Observation:

- The relationships among numerical features appear weak.
- No clear clusters are visible.

9 Summary of Findings

1. The dataset contains mostly recent content.
2. Movies are more common than TV Shows.
3. Numerical features show weak correlations.
4. A few outliers are present in the data.
5. No strong relationships were identified between the selected variables.
6. Overall, the dataset is suitable for exploratory analysis and visualization.